

$$\frac{105}{33} = a + \frac{1}{b + \frac{1}{c + \frac{1}{d + \frac{1}{e + \frac{1}{f + \frac{1}{g + \frac{1}{\dots}}}}}}}$$

where the ellipses could be referring to a finite rather than an infinite continuation. Our algorithm will generate a first, then b , then c , and proceed through terms of the alphabet sequentially until it reaches the final term or until we require it to stop.

If we interpret our example $105/33$ as a division problem instead of a fraction, we find that $105/33$ is 3, remainder 6. We can rewrite $105/33$ as $3 + 6/33$:

$$3 + \frac{6}{33} = a + \frac{1}{b + \frac{1}{c + \frac{1}{d + \frac{1}{e + \frac{1}{f + \frac{1}{g + \frac{1}{\dots}}}}}}}$$

The left and the right sides of this equation both consist of an integer (3 and a) and a fraction ($6/33$ and the rest of the right side). We conclude that the integer parts are equal, so $a = 3$. After this, we have to find a suitable b , c , and so on such that the whole fractional part of the expression will evaluate to $6/33$.

To find the right b , c , and the rest, look at what we have to solve after concluding that $a = 3$: